ORGANIZATION
DIGITAL TECHNOLOGIES
SENSORS
ACTUATORS
COMPUTER VISION
LARGE LANGUAGE MODELS
USER INTERFACES
NATURAL LANGUAGE PROCESSING
CLOUD SERVICES
DATABASES

The slides are meant as visual support for the lecture.
They are neither a documentation nor a script.

Please do not print the slides.

Comments and feedback at n.meseth@hs-osnabrueck.de

ORGANIZATION

ILIAS
Microsoft Teams

sessions

group work

examination

working environment

visual studio code
python
tinkerforge
git

DIGITAL TECHNOLOGIES

a model for solving problems

cyber physical systems

artificial intelligence

software prototyping

cyber physical systems

sensors

actuators

artificial intelligence

software prototyping

temperature
humidity
co2
uv light
ambient light
sound pressure
thermal image
camera
...

led
speaker
display
motor
…

cyber physical systems

artificial intelligence

software prototyping

computer vision

generative ai

natural language processing

cyber physical systems

artificial intelligence

software prototyping

computer vision

generative ai

natural language processing

image classification
image segmentation
object recognition
object tracking
face recognition
face identification
emotion recognition
pose estimation
text recognition

cyber physical systems

artificial intelligence

software prototyping

computer vision

generative ai

natural language processing

text generation
text summary
text analysis
image generation
image description
video generation
music generation

cyber physical systems

artificial intelligence

software prototyping

computer vision

generative ai

natural language processing

speech-to-text
text-to-speech
translation

cyber physical systems

artificial intelligence

software prototyping

user interfaces

cloud services

databases

introductory example

visual studio code
programs
python

LEDs

large language models

speech-to-text

user interface

SENSORS

temperature / humidity

rgb led button

camera

thermal imaging camera

microphone

keyboard

temperature / humidity

```
th = BrickletHumidityV2(UID, ipcon)...
```

```
th.get_humidity()
th.get_temperature()
```

```
th.register_callback(th.CALLBACK_HUMIDITY, cb_humidity)
th.register_callback(th.CALLBACK_TEMPERATURE, ...)
```

```
th.set_humidity_callback_configuration(250, False, "x", 0, 0)
th.set_temperature_callback_configuration(...)
```

rgb led button

```
btn = BrickletRGBLEDButton(UID, ipcon)...
```

```
btn.set_color(255, 0, 0)
```

```
btn.get_button_state()
```

```
btn.register_callback(...)
```

camera

OpenCV

```
import cv2
```

```
# Get video capture device (webcam)
webcam = cv2.VideoCapture(0)
```

```
# Read a frame
success, frame = webcam.read()
```

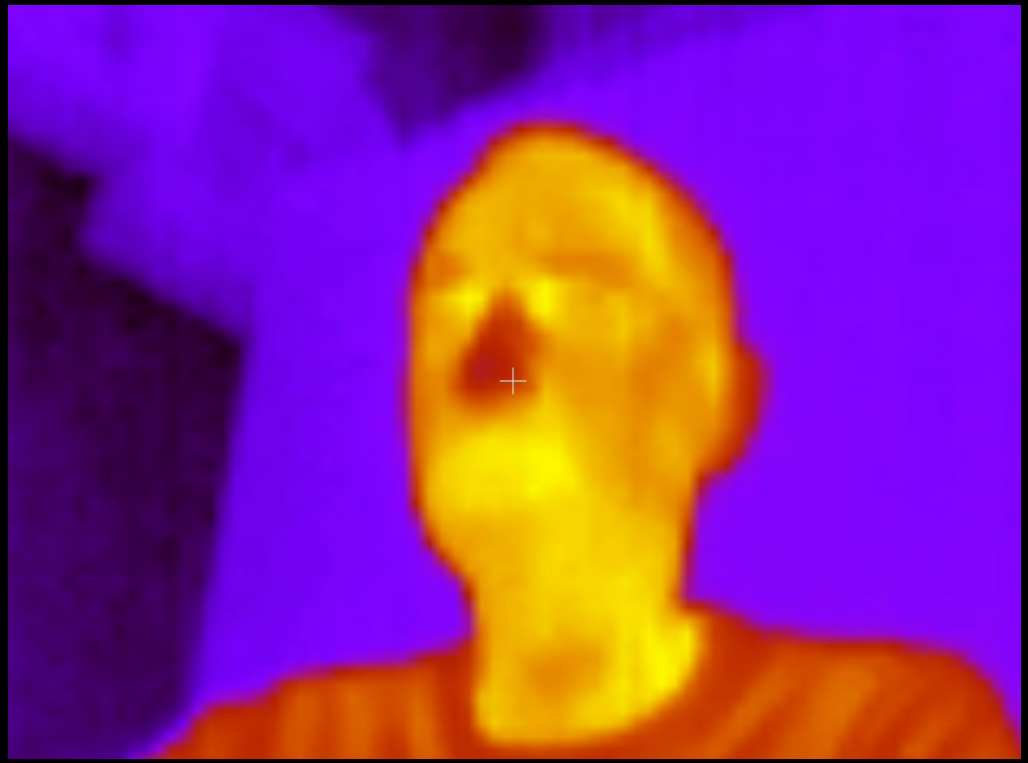
```
# Show the image from the frame
cv2.imshow("Webcam", frame)
```

```
# Save the frame as .png
cv2.imwrite("screenshot.png", frame)
```

thermal imaging camera

OpenCV

Tinkerforge



```
ti = BrickletThermalImaging(UID, ipcon)
ti.set_image_transfer_config(...)
img = ti.get_high_contrast_image()
```

```
ti.register_callback(...)
```

microphone

```
import pyaudio
```

```
# Define recording parameters
FORMAT = pyaudio.paInt16
CHANNELS = 1
RATE = 44100
CHUNK = 1024
```

```
# Get access to the microphone
audio = pyaudio.PyAudio()
```

```
# Start listening
stream = audio.open(...)
```

```
# Read a chunk of frames
stream.read(CHUNK)
```

```
# Stop and close stream
stream.stop_stream()
stream.close()
```

```
# Terminate access to microphone
audio.terminate()
```

keyboard

```
import keyboard
```

```
# Define a callback function for a key
def record_audio():
    print("Recording audio...")
```

```
# Add key listener
keyboard.add_hotkey("r", record_audio)
```

```
# Wait until a specific key was pressed
keyboard.wait("esc")
```

ACTUATORS

rgb led

OLED display

speaker

rgb led

```
led = BrickletRGBLEDV2(UID, ipcon)
led.set_rgb_value(255, 0, 0)
```

OLED display

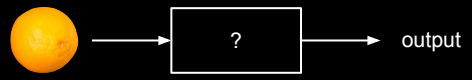
```
oled = BrickletOLED128x64V2(UID, ipcon)
oled.clear_display()
oled.write_line(0, 0, "Welcome!")
```

speaker

```
import simpleaudio as sa
```

```
# Create a wave object from .wav-file and play it
wav = sa.WaveObject.from_wave_file("sound.wav")
wav.play().wait_done()
```

COMPUTER VISION



finding images with oranges

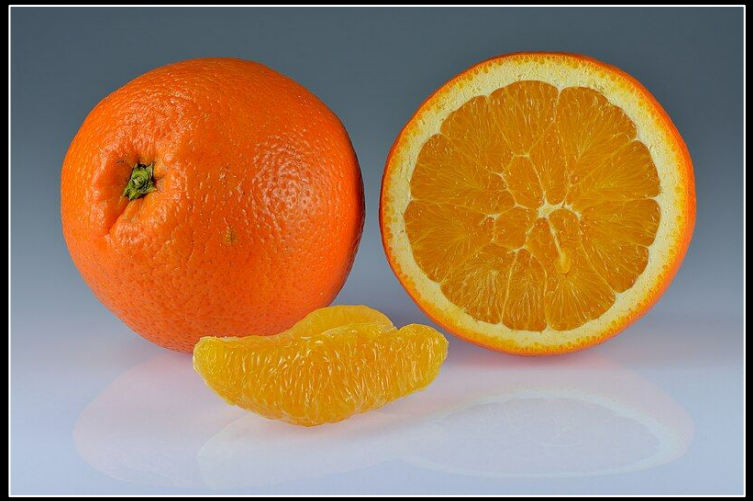


Image source: Wikimedia

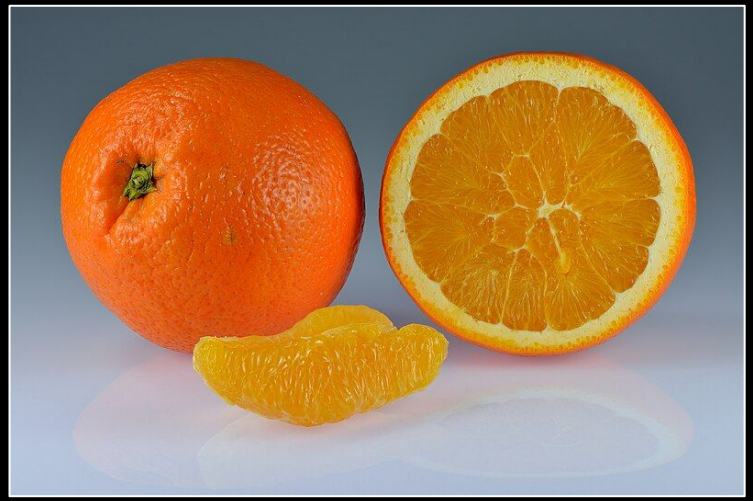


Image source: Wikimedia

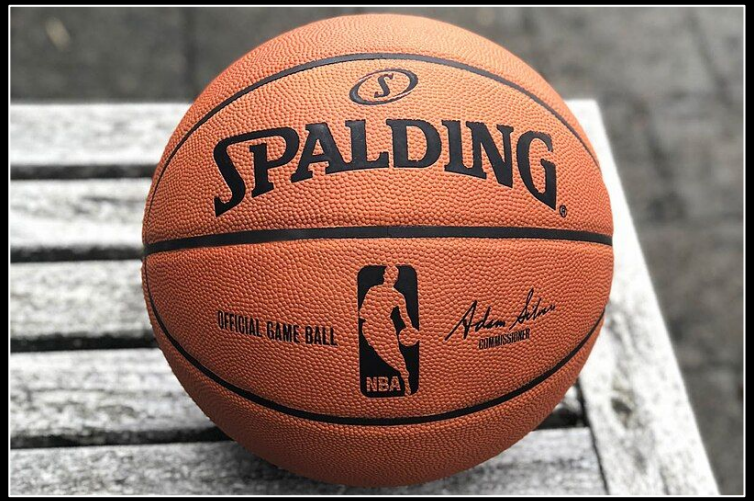
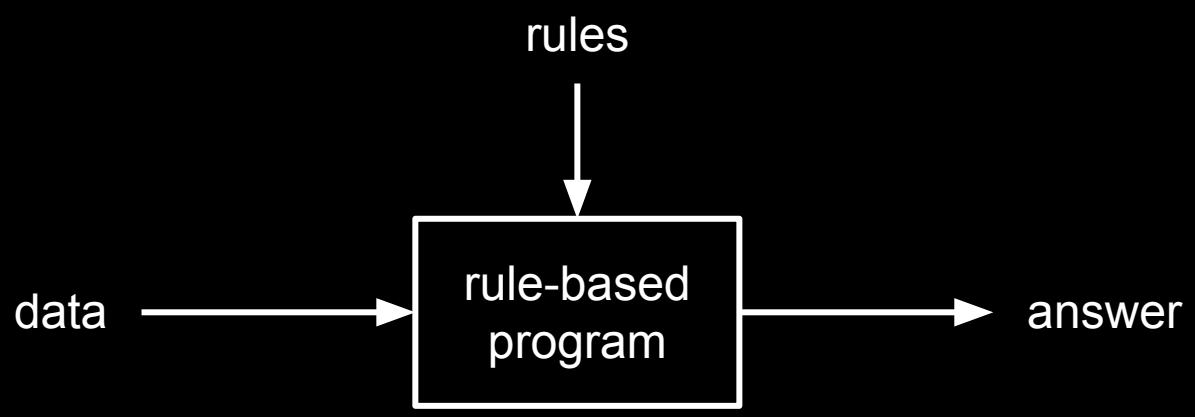
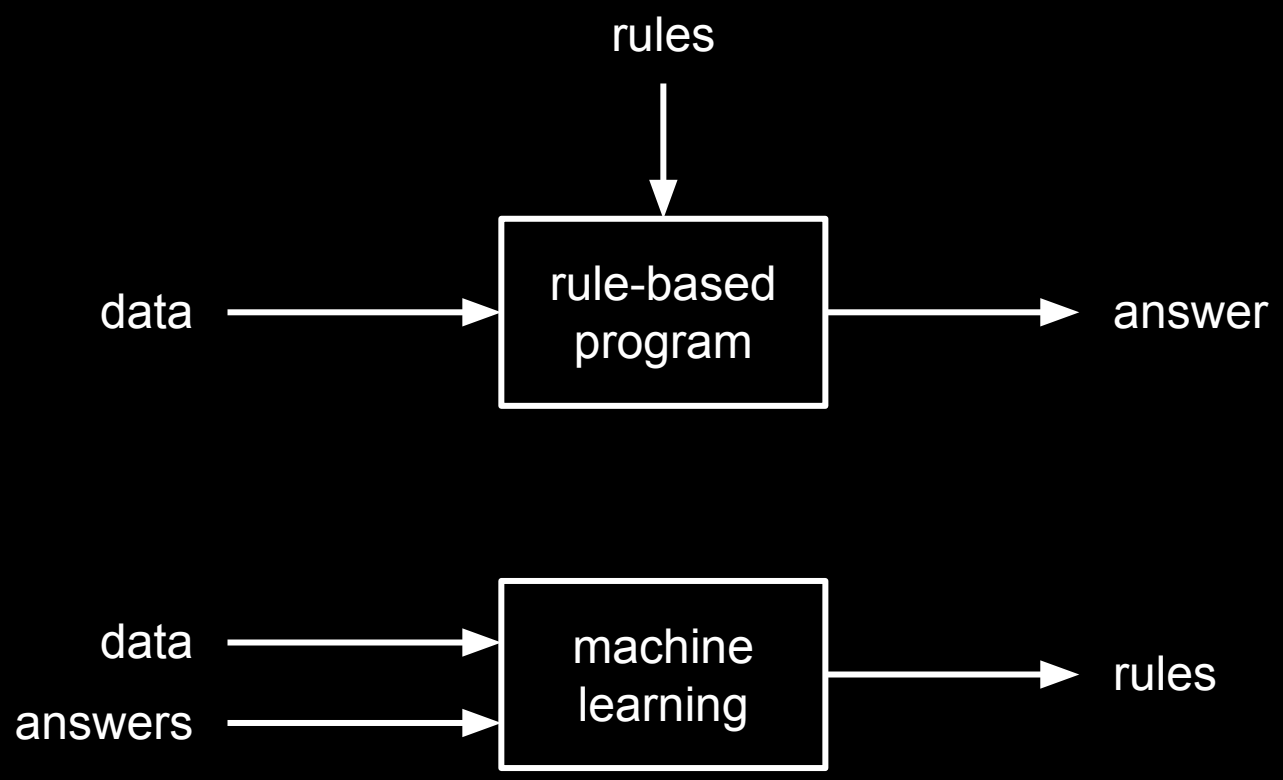


Image source: Wikimedia

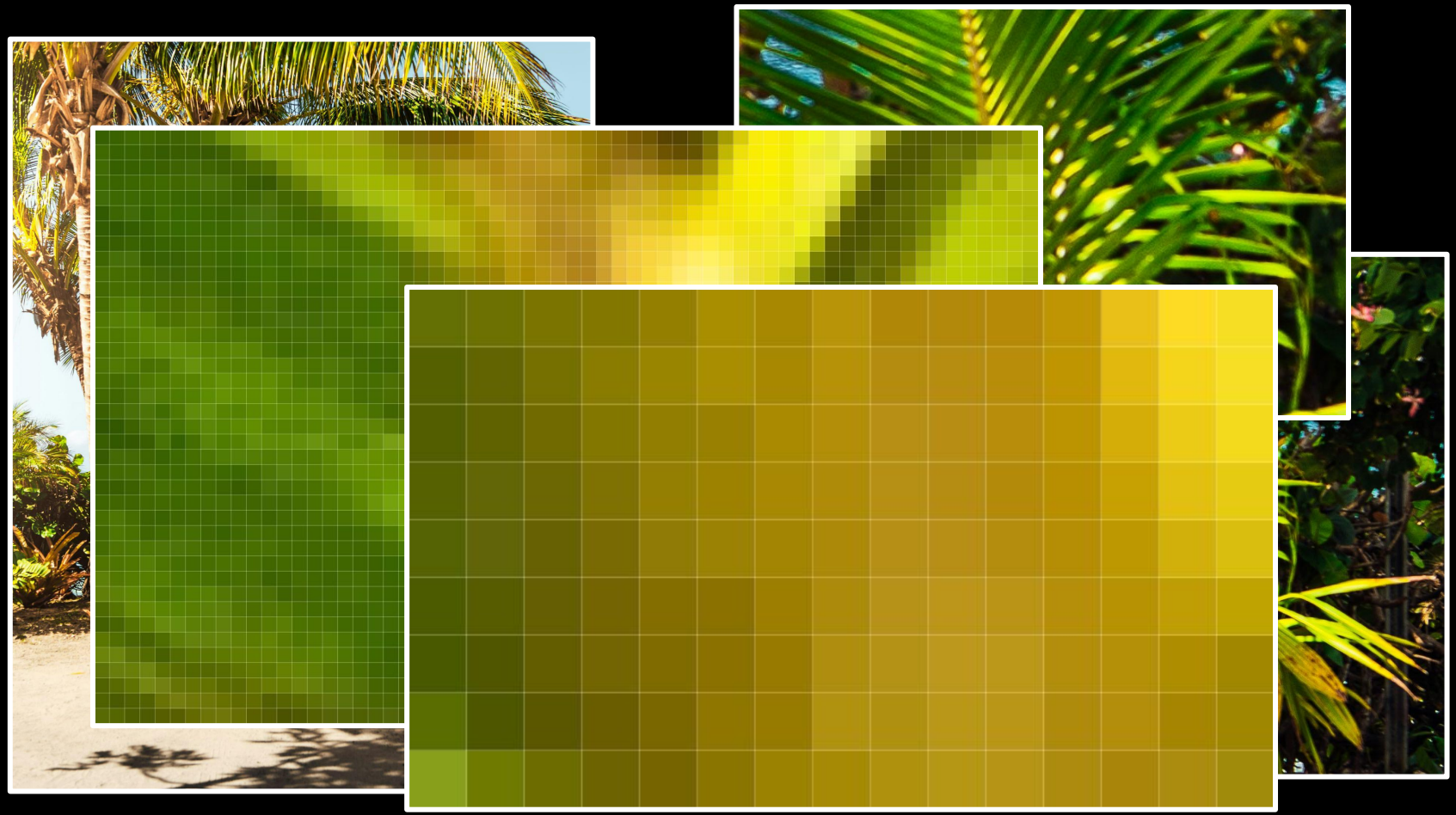
what set of rules can solve this?

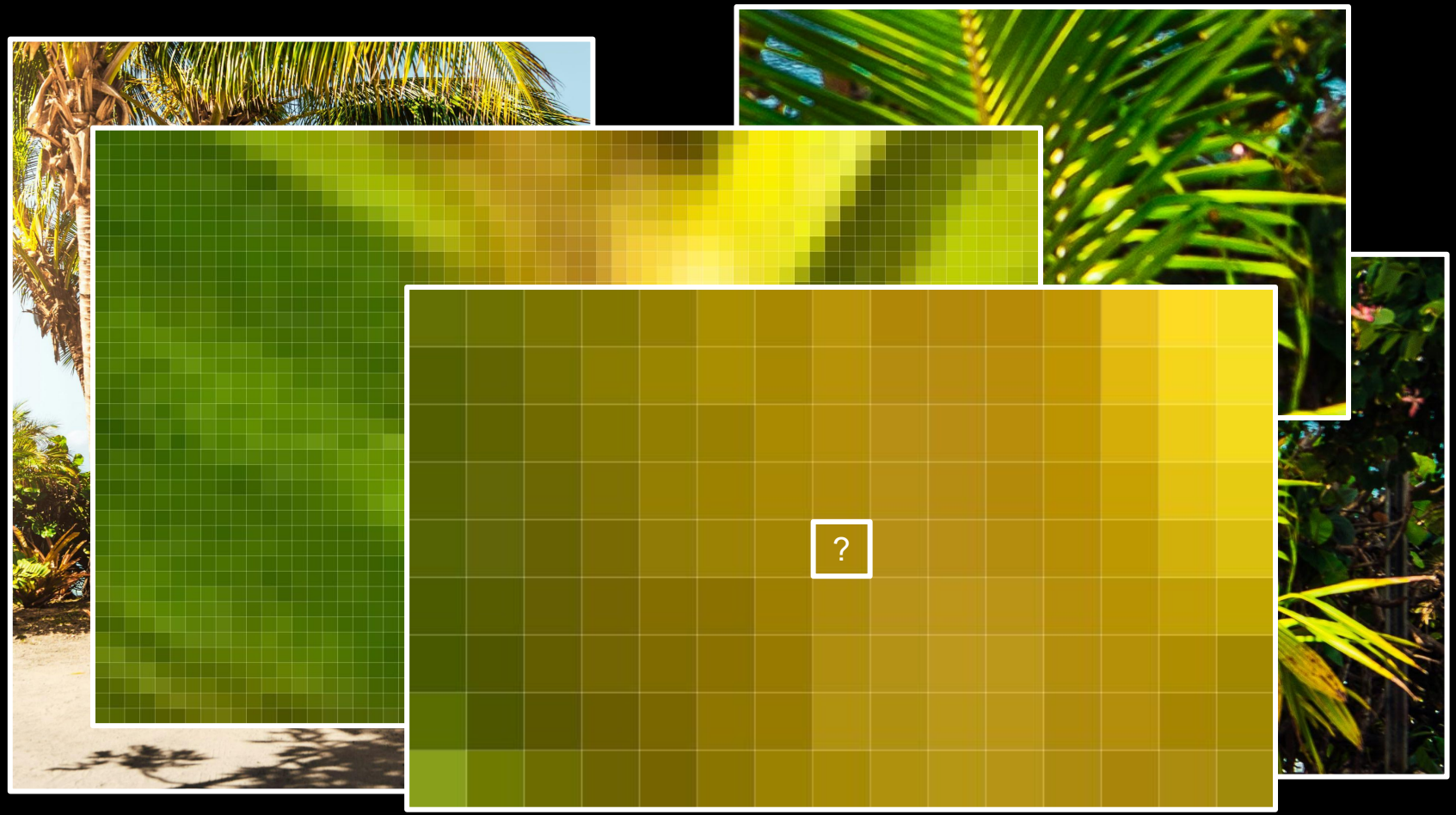
machine learning algorithms

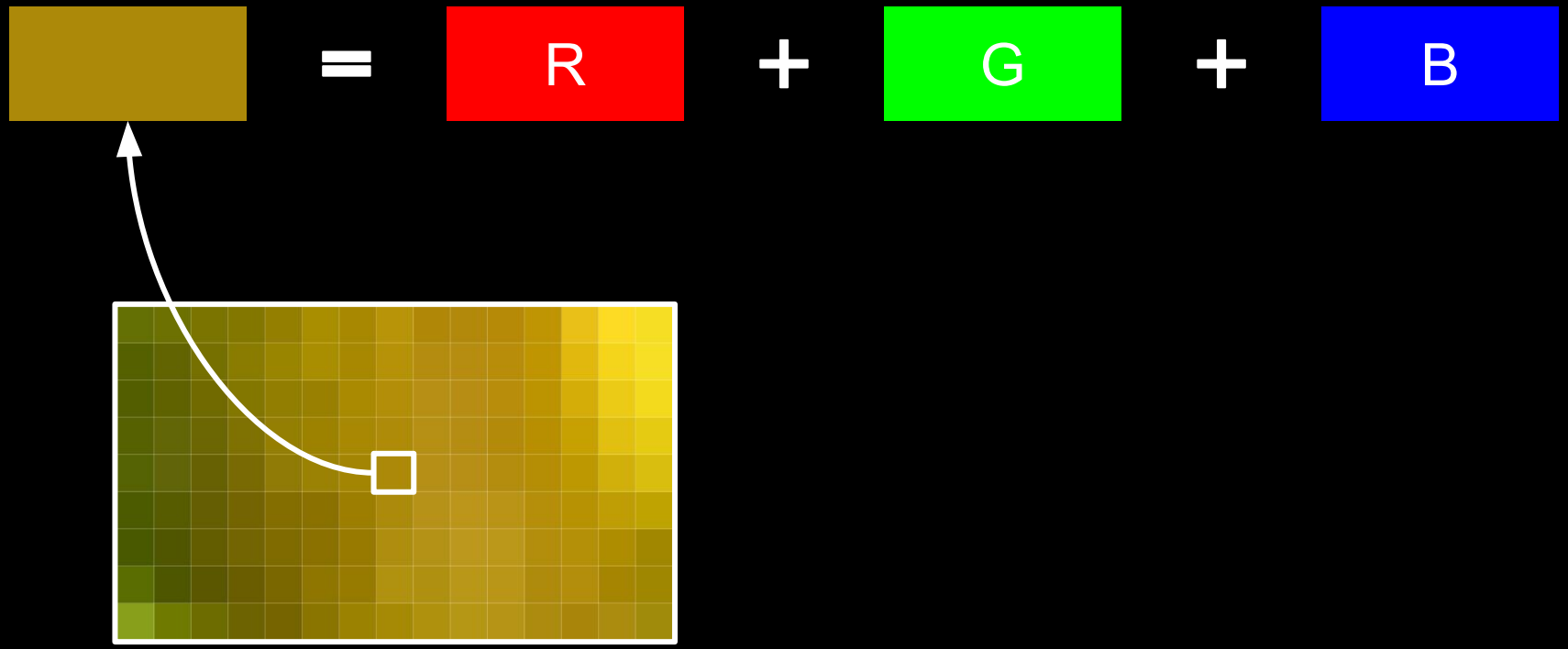

rules
data
rule-based
program
answer


rules
data
rule-based program
answer
data
answers
machine learning
rules

images in a computer






=
R
+
G
+
B

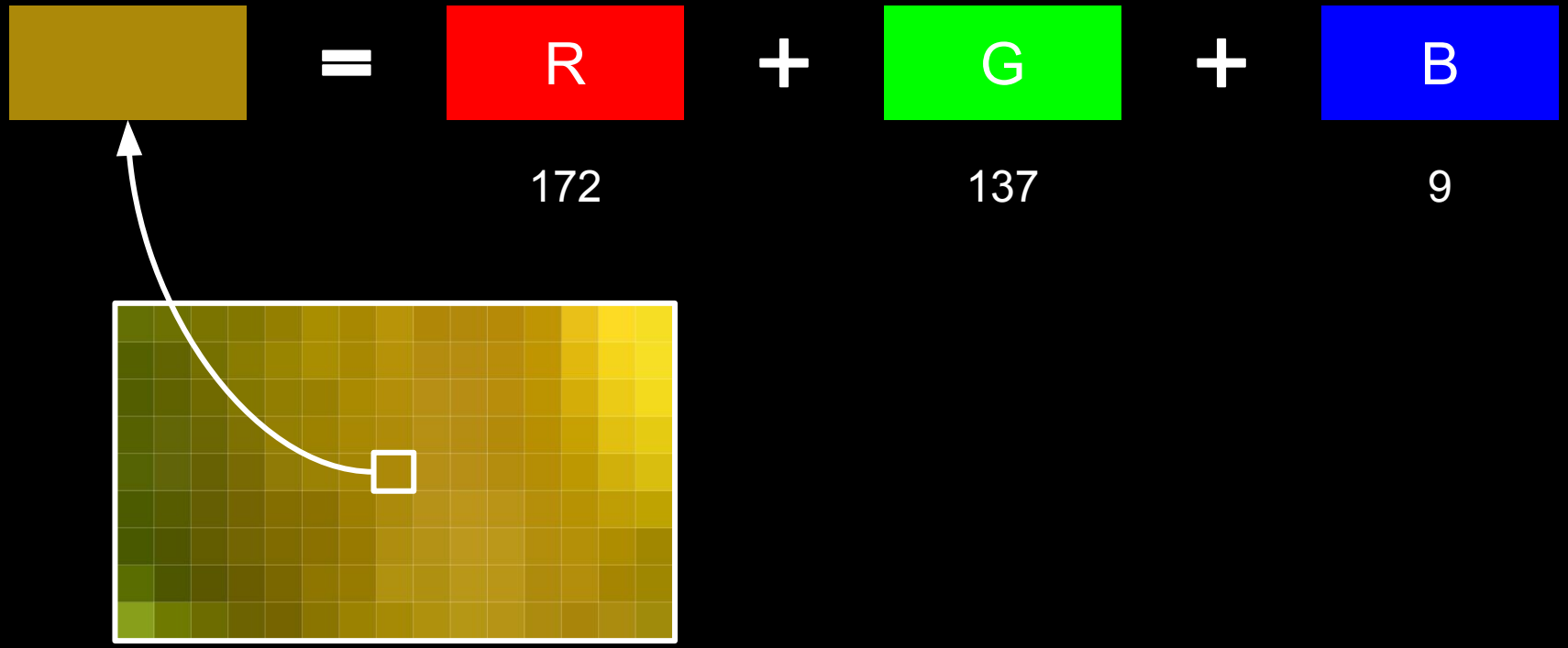

=
R
+
G
+
B
172
137
9

image classification

Q: Does an image belong to one or the other class from a fixed set of classes?

Cat or Dog?

model

"cat"

Cat or Dog?

model → "cat"

model → "dog"

Google's teachable machine

https://teachablemachine.withgoogle.com

```
pip install keras
pip install tensorflow==2.12.0
```

```
# Load the classifier and class names
model = load_model("my_model.h5")
class_names = open("labels.txt", "r").readlines()
```

```
# Convert the image t0 224 x 224
image = cv2.resize(image, (224, 224), interpolation=cv2.INTER_AREA)

# Turn into a list of pixels
image = np.asarray(image, dtype=np.float32).reshape(1, 224, 224, 3)

# Normalize each pixel's color value (-1/1)
image = (image / 127.5) - 1
```

```
# Make a prediction for the class
prediction = model.predict(image)

# Get the class with the highest confidence value
index = np.argmax(prediction)
class_name = class_names[index]

# Get the confidence score for the predicted class
confidence_score = prediction[0][index]
```

Class: Labeler
Confidence: 1.00
brother

YOLO v8 Image Classification

https://docs.ultralytics.com/

```
pip install ultralytics
```

```python
# Load the classifier
from ultralytics import YOLO
model = YOLO("yolov8n-cls.pt")
```

```
# Make a prediction
results = model('cat.jpg')
```

```
# Show result
results[0].show()
```

```python
# Get the top result
top = results[0].probs.top1
class_name = results[0].names[top]
print(class_name)
```

zero-shot image classification

Q: Which classes do you train your model on?

GPT-4 Vision

```
pip install openai
```

```
# import openai API and set api key
from openai import OpenAI
os.environ["OPENAI_API_KEY"] = "..."
client = OpenAI()
```

```
# define a suitable prompt for the task
prompt = "Classify the image into 'dog' or 'cat'. Return
only the word for the class of the image."
```

```python
# This function is needed to encode an image to base64 for OpenAI's API
def encode_image(image_path):
  with open(image_path, "rb") as image_file:
    return base64.b64encode(image_file.read()).decode('utf-8')

image_path = "cat.webp"
image = encode_image(image_path)
```

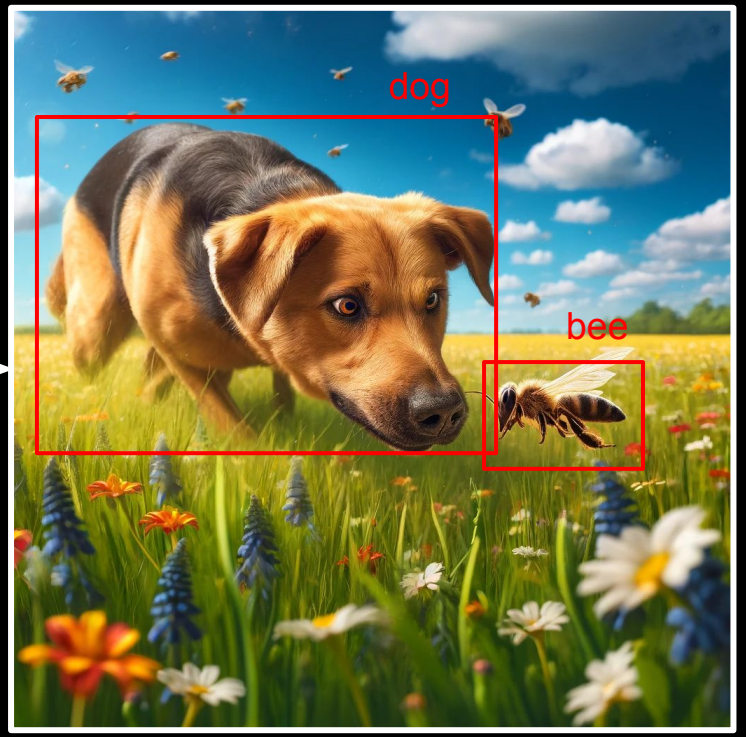
```
response = client.chat.completions.create(
  model="gpt-4-turbo",
  messages = [
    { "role": "user", "content": [
        { "type": "text", "text": prompt },
        { "type": "image_url", "image_url": { "url": f"{image}" } }
      ]
    }
  ],
  max_tokens=300,
)
```

```
# Show the answer of the classification
print(response.choices[0].message.content)
```

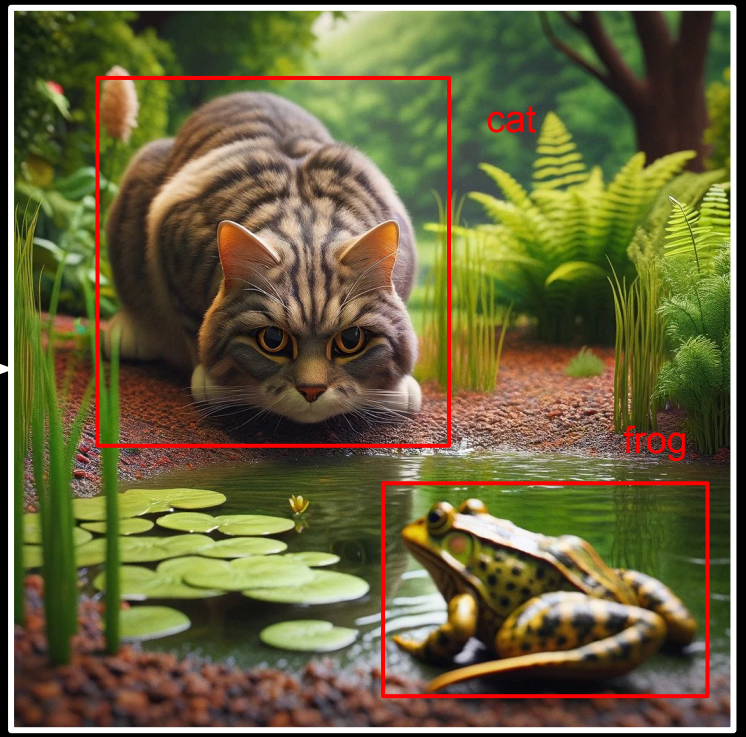
object detection

Q: Which objects are in the image and where?

AI



AI



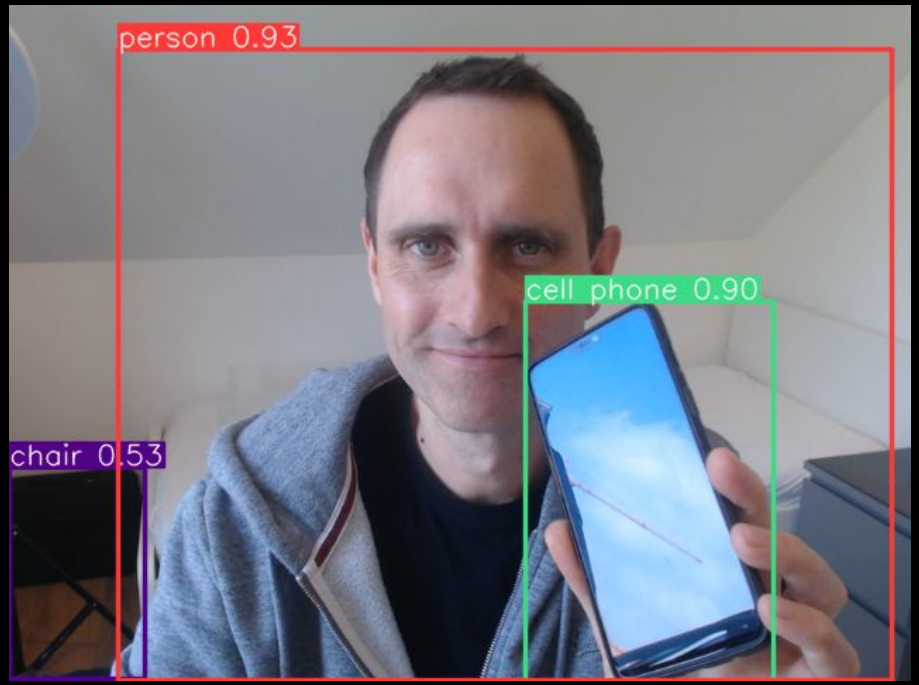
YOLO v8 Object Detection

https://docs.ultralytics.com/

```python
# Load the detector
from ultralytics import YOLO
model = YOLO("yolov8n.pt")
```

```
# Make a prediction one each frame
results = model(frame)

# Annotate frame
annotated_frame = results[0].plot()
```


person 0.93
cell phone 0.90
chair 0.53

Q: Which objects do you teach your model to recognize?

zero-shot object detection

"Simple Open-Vocabulary Object Detection with Vision Transformers"

https://arxiv.org/abs/2205.06230

```
# Load the open world detector
from ultralytics import YOLO
model = YOLO("yolov8s-world.pt")
```

```
# Define custom objects to look for
model.set_classes(["person with glasses"])
```

```
# Make a prediction one each frame
results = model(frame)

# Annotate frame
annotated_frame = results[0].plot()
```

optical character recognition (OCR)

Getränke
HOFFMANN

B. Bobzin
Bramscher Straße 159
49088 O S N A B R Ü C K
Mo-Fr.08:00-19:30 Uhr Sa.08:00-19:00 Uhr
Tel.0541/684726

26.04.24 09:07 2347 00002 01 #306521

6x 20er KASTEN á 13.29
#133075 Salvus Apfelschorle 0,33 79.74 1
#000901 Pfandflasche 120x0.15 18.00 1
#000905 Leerkiste 6x1.50 9.00 1

8x 20er KASTEN á 7.79
#133734 Salvus mit Kribbel 0,33L 62.32 1
#000901 Pfandflasche 160x0.15 24.00 1
#000905 Leerkiste 8x1.50 12.00 1

Endsumme € 205.06

davon Ware EUR : 142.06
davon Pfand EUR : 63.00
abzgl. Rückpfand EUR : 0.00

MwSteuer	Netto	MwSt.	Brutto
19.00%	172.32	32.74	205.06 1

Kartenzahlung EUR : 205.06
Kartenart : Visa Debit
BelegNr : 6988
PAN : ##########07641
Kartenfolgenr. : 0000
VU-Nr. : 228165299
zurück EUR : 0.00

******* Keine P A Y B A C K Karte? ******
Dein PAYBACK Vorteil für diesen Einkauf
wären 71°P gewesen!
Hier PAYBACK Karte mitnehmen oder auf
getraenke-hoffmann.de/payback anmelden
****

GH-St.-Nr.: 047 225 19041 (#)
Vielen Dank
für Ihren Einkauf!

tesseract

GPT-4 Vision

```
# define a suitable prompt for the task
prompt = "Extract all food and beverage items with their quantity and price from this receipt into a JSON list. The receipt is in German."
```

```
response = client.chat.completions.create(
  model="gpt-4o",
  response_format={ "type": "json_object" },
  messages = [
    { "role": "user", "content": [
        { "type": "text", "text": prompt },
        { "type": "image_url", "image_url": { "url": f"{image}" } }
      ]
    }
  ],
  max_tokens=300,
)
```

LARGE LANGUAGE MODELS

what has been said so far?

(*history* + *prompt*)

what has been said so far?
(*history* + *prompt*)

prediction of next token based on
learnt probability distribution

what has been said so far?
(*history* + *prompt*)

prediction of next token based on
learnt probability distribution

+

(randomness)

what has been said so far?
(*history* + *prompt*)

prediction of next token based on learnt probability distribution

+

(randomness)

+

(filter)

(discriminating, insulting content)

what has been said so far?
(*history* + *prompt*)

prediction of next token based on learnt probability distribution

+

(randomness)

+

(filter)

(discriminating, insulting content)

next word (*token*)

what has been said so far?
(*history* + *prompt*)

prediction of next token based on
learnt probability distribution

+

(randomness)

+

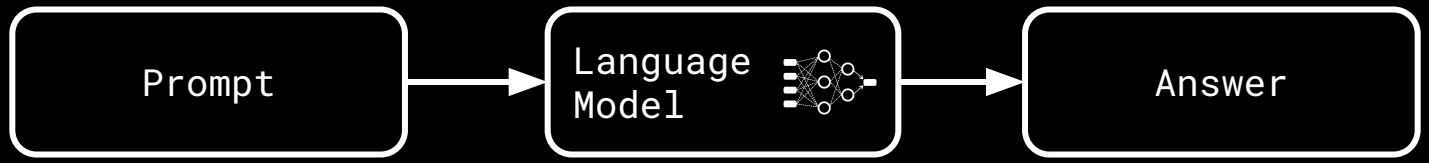
(filter)

(discriminating, insulting content)

next word (*token*)

PROMPTING

https://www.promptingguide.ai/


Prompt
Language Model
Answer

elements of a prompt

```
<instruction>
<context>
<input data>
<output indicator>
```

elements of a prompt	example prompt
`<instruction>`	`Explain the binary number system.`
`<context>`	
`<input data>`	
`<output indicator>`	

elements of a prompt

<instruction>

<context>

<input data>

<output indicator>

example prompt

Explain the binary number system.

start simple

elements of a prompt

<instruction>

<context>

<input data>

<output indicator>

example prompt

You are a friendly tutor and your task is to explain complex concepts as simple as possible.

Explain the binary number system.

elements of a prompt

<instruction>

<context>

<input data>

<output indicator>

example prompt

You are a friendly tutor and your task is to explain complex concepts as simple as possible.

Your answers are never longer than 10 sentence.

Explain the binary number system.

ZERO-SHOT PROMPTING

elements of a prompt	example prompt
<instruction>	Classify the text into neutral, negative or positive.
<context>	
<input data>	Text: "What a great dinner!"
<output indicator>	Sentiment:

elements of a prompt

<instruction>

<context>

<input data>

<output indicator>

example prompt

```
Classify the text into neutral,
negative or positive.

Text: "What a great dinner!"

Sentiment:
```

this will be replaced with data later…

FEW-SHOT PROMPTING

IN-CONTEXT LEARNING

examples in the context to learn from

```
Extract all references to countries and their continent in the following text
using the format from the examples below.

Example 1: "They played the team called 'Die Mannschaft' in the world cup final"
Correct answer: Germany, Europe

Example 2: "The Three Lions once again lost to Germany in a semi final"
Correct answer: England, Europe, Germany, Europe

Text: "The Selecao was destroyed 1:7 by the DFB selection in their home stadium."
Answer:
```

examples in the context to learn from

Extract all references to countries and their continent in the following text using the format from the examples below.

Example 1: "They played the team called 'Die Mannschaft' in the world cup final"
Correct answer: Germany, Europe

Example 2: "The Three Lions once again lost to Germany in a semi final"
Correct answer: England, Europe, Germany, Europe

Text: "The Selecao was destroyed 1:7 by the DFB selection in their home stadium."
Answer:

more prompting strategies

chain-of-thought (CoT)
self-consistency
generate knowledge prompting
prompt chaining (subtasks)
tree-of-thoughts (ToT)
retrieval-augmented-generation (RAG)
…

OpenAI

```
pip install openai
```

```python
from openai import OpenAI
import os

os.environ["OPENAI_API_KEY"] = "<YOUR_API_KEY>"
client = OpenAI()
```

```
# define a system message
system_message = """
     You are a world-famous 5-star chef. Based on ingredients the user has at home,
     you suggest easy-to-cook recipes. """
```

```
# define a prompt for the task
prompt = """
     Suggest a recipe for lunch.

     List of ingredients:
     - butter
     - eggs
     - flour
     - salt
     - milk

     Recipe: """
```

```
response = client.chat.completions.create(
  model="gpt-4o",
  messages = [
    {"role": "system", "content": system_message },
    {"role": "user", "content":  prompt },]
    }
  ],
  max_tokens=2000
)
```

USER INTERFACES

streamlit

https://docs.streamlit.io/ # official documentation
https://streamlit.io/components # third-party extensions

```
pip install streamlit
```

```
- Home.py
- pages/
    - 1_Speech.py
    - 2_Webcam.py
    - 3_Microphone.py
- lib/
    - speech_to_text.py
    - text_to_speech.py
    - vision.py
```

entry point to our UI

```
- Home.py
- pages/
    - 1_Speech.py
    - 2_Webcam.py
    - 3_Microphone.py
- lib/
    - speech_to_text.py
    - text_to_speech.py
    - vision.py
```

entry point to our UI

more pages in our app

```
- Home.py
- pages/
    - 1_Speech.py
    - 2_Webcam.py
    - 3_Microphone.py
- lib/
    - speech_to_text.py
    - text_to_speech.py
    - vision.py
```

```
- Home.py
- pages/
    - 1_Speech.py
    - 2_Webcam.py
    - 3_Microphone.py
- lib/
    - speech_to_text.py
    - text_to_speech.py
    - vision.py
```

entry point to our UI

more pages in our app

our custom functions we'd like to use from our UI

Home.py

```
import streamlit as st

st.title("My first UI")
st.write("This is a simple UI for prototyping our application.")

name = st.text_input("Enter your name")

if st.button("Greet me"):
    st.write(f"Hello {name} 🤞")
```

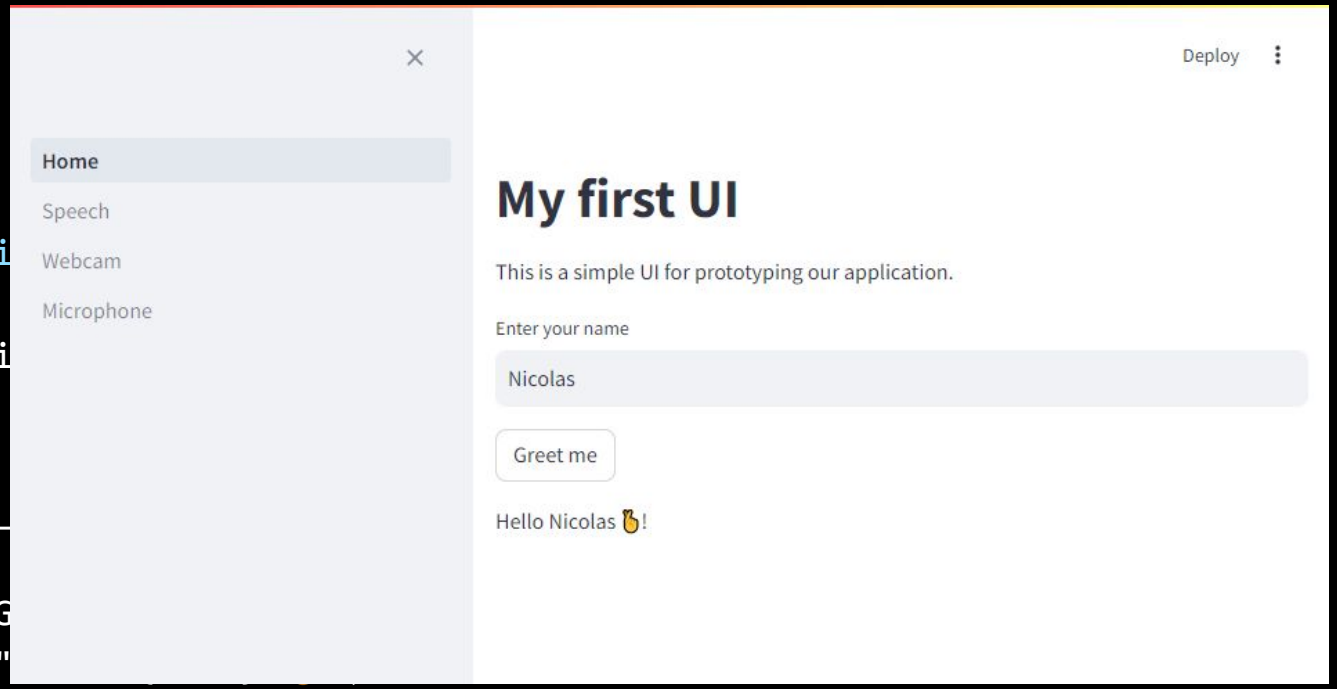
Home.py

```
import streamli

st.title("My fi
st.write("This

name = st.text_

if st.button("G
    st.write(f"
```



1_Speech.py

```
import streamlit as st
from pages.lib.text_to_speech import text_to_speech

st.title("Speech demo")
st.write("Enter a text and it will be converted to speech.")

text = st.text_input("Enter some text")
voice = st.selectbox("Select a voice", ["alloy", … "shimmer"])

if st.button("Turn to speech"):
    audio_file = text_to_speech(text, voice=voice)
    st.audio(audio_file.as_posix(), format="audio/mpeg")
```

1_Speech.py

```
import streamlit as st
from pages.lib.text_to_speech import text_to_speech

st.title("Speech demo")
st.write("Enter a text and it will be converted to speech.")

text = st.text_input("Enter some text")
voice = st.selectbox("Select a voice", ["alloy", … "shimmer"])

if st.button("Turn to speech"):
    audio_file = text_to_speech(text, voice=voice)
    st.audio(audio_file.as_posix(), format="audio/mpeg")
```

from lib/text_to_speech.py

lib/text_to_speech.py

```
from openai import OpenAI
import os
os.environ["OPENAI_API_KEY"] = "..."
client = OpenAI()

def text_to_speech(text, voice="alloy"):
    speech_file_path = Path(__file__).parent / "speech.mp3"
    response = client.audio.speech.create(
        model="tts-1",
        voice=voice,
        input=text
    )

    response.write_to_file(speech_file_path)
    return speech_file_path
```

lib/text_to_speech.py

```
from openai import OpenAI
import os
os.environ["OPENAI_API_KEY"] = "..."
client = OpenAI()
```

setup OpenAI API

```
def text_to_speech(text, voice="alloy"):
    speech_file_path = Path(__file__).parent / "speech.mp3"
    response = client.audio.speech.create(
        model="tts-1",
        voice=voice,
        input=text
    )

    response.write_to_file(speech_file_path)
    return speech_file_path
```

lib/text_to_speech.py

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from openai import OpenAI
import os
os.environ["OPENAI_API_KEY"] = "..."
client = OpenAI()
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setup OpenAI API

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def text_to_speech(text, voice="alloy"):
    speech_file_path = Path(__file__).parent / "speech.mp3"
    response = client.audio.speech.create(
        model="tts-1",
        voice=voice,
        input=text
    )

    response.write_to_file(speech_file_path)
    return speech_file_path
```

define custom function

2_Webcam.py

```
import streamlit as st
from pages.lib.vision import ask_gpt4o


st.title("Video camera test")

picture = st.camera_input("Take a picture")

if picture:
    st.image(picture)
    answer = ask_gpt4o("What is in this picture?", picture)
    st.write(answer)
```

2_Webcam.py

```
import streamlit as st
from pages.lib.vision import ask_gpt4o
```

from lib/vision.py

```
st.title("Video camera test")

picture = st.camera_input("Take a picture")

if picture:
    st.image(picture)
    answer = ask_gpt4o("What is in this picture?", picture)
    st.write(answer)
```

lib/vision.py

```
from openai import OpenAI
import os
os.environ["OPENAI_API_KEY"] = "..."
client = OpenAI()

def encode_image(image_buffer):

def ask_gpt4o(prompt, image_buffer):
  image = encode_image(image_buffer)
  response = client.chat.completions.create(
    model="gpt-4o",
    messages=[
      {
        "role": "user", "content": [
          { "type": "text", "text": prompt },
          { "type": "image_url", "image_url": { "url": f"{image}" } }
        ],
      }
    ]
    )

  return response.choices[0].message.content
```

lib/vision.py

```
from openai import OpenAI
import os
os.environ["OPENAI_API_KEY"] = "..."
client = OpenAI()
```

setup OpenAI API

```
def encode_image(image_buffer):
```

```
def ask_gpt4o(prompt, image_buffer):
  image = encode_image(image_buffer)
  response = client.chat.completions.create(
    model="gpt-4o",
    messages=[
      {
        "role": "user", "content": [
          { "type": "text", "text": prompt },
          { "type": "image_url", "image_url": { "url": f"{image}" } }
        ],
      }
    ]
    )

  return response.choices[0].message.content
```

define custom function

3_Microphone.py

```
import streamlit as st
from streamlit_mic_recorder import mic_recorder
from pages.lib.text_to_speech import speech_to_text

st.title("Microphone test")

def callback():
    if st.session_state.my_recorder_output:
        audio = st.session_state.my_recorder_output
        text = text_to_speech(audio)
        st.success(text)

audio = mic_recorder(key='my_recorder', callback=callback)
```

3_Microphone.py

pip install streamlit-mic-recorder

```
import streamlit as st
from streamlit_mic_recorder import mic_recorder
from pages.lib.text_to_speech import speech_to_text

st.title("Microphone test")

def callback():
    if st.session_state.my_recorder_output:
        audio = st.session_state.my_recorder_output
        text = text_to_speech(audio)
        st.success(text)

audio = mic_recorder(key='my_recorder', callback=callback)
```

3_Microphone.py

pip install streamlit-mic-recorder

```
import streamlit as st
from streamlit_mic_recorder import mic_recorder
from pages.lib.text_to_speech import speech_to_text

st.title("Microphone test")

def callback():
    if st.session_state.my_recorder_output:
        audio = st.session_state.my_recorder_output
        text = text_to_speech(audio)
        st.success(text)

audio = mic_recorder(key='my_recorder', callback=callback)
```

from lib/speech_to_text.py

lib/speech_to_text.py

```
from openai import OpenAI
import os
import io

os.environ["OPENAI_API_KEY"] = "..."
client = OpenAI()

def speech_to_text(audio):
    audio_bio = io.BytesIO(audio['bytes'])
    audio_bio.name = 'audio.mp3'

    transcription = client.audio.transcriptions.create(
        model="whisper-1",
        file=audio_bio
    )
    return transcription.text
```

lib/speech_to_text.py

```
from openai import OpenAI
import os
import io

os.environ["OPENAI_API_KEY"] = "..."
client = OpenAI()
```

setup OpenAI API

```
def speech_to_text(audio):
    audio_bio = io.BytesIO(audio['bytes'])
    audio_bio.name = 'audio.mp3'

    transcription = client.audio.transcriptions.create(
        model="whisper-1",
        file=audio_bio
    )
    return transcription.text
```

define custom function